Walnut Technologies

132, JLPL Industrial Area, Sec 82, Mohali –Punjab India ,
Email: contact@walnutmedical.in +91-9650622772

Product Brochure
Soundbox

Walnut powered smart sound enables you to receive real time audio payment confirmations while accepting business payments via QR codes from any UPI enabled app; configurable for global standards and deployments.



Front Side



Back side

Soundbox Specification :

Category	Item	Specifications
Hardware	Module	High performance Dual core ARM CortexR
Hardware	Memory	8MB
Hardware	Communication	4G (LTE Cat1)
Hardware	Bands Supported	LTE FDD: B1/B3/B5/B8, LTE TDD: B34/B38/B39/B40/B41
Hardware	Indicator	1*RGB LED
Hardware	Speaker	Nominal 2W, Peak 3W > (84-87)dB sound output
Hardware	Audio	PA: 3.2W@5V/4Ω/10% THD
Hardware	Key	4*Keys+1*Reset Key
Hardware	Battery	2000mAh
Hardware	USB Port	Type C
Hardware	SIM Tray Type	Nano
Hardware	Shell	Material: ABS, Fire rating: HB, Drop Test: 1.0M
Software	Remote updates	APP, language & Firmware Update
Software	Secure Connectivity	Support SSL/TLS
Accessories	Ac-DC Adapter	AC 90-300V, DC 5V/1A (Optional)
Accessories	USB Cable	1 mtr USB Type C, Supports 1A Charging
Accessories	Certification	BIS

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Product Brochure

DQR 4.3” with Keypad



Front Side



Back Side

Soundbox Specification :

Category	Item	Specifications
Hardware	Module	High performance Dual core ARM CortexR
Hardware	Memory	8MB
Hardware	Communication	4G (LTE Cat1) or 4G + 2G support (Optional)
Hardware	Bands Supported	LTE FDD: B1/B3/B5/B8, LTE TDD: B34/B38/B39/B40/B41
Hardware	Indicator	1*RGB LED
Hardware	Speaker	Nominal 2W, Peak 3W > (84-87)dB sound output
Hardware	Audio	PA: 3.2W@5V/4Ω/10% THD
Hardware	Key	4*Keys+1*Reset Key
Hardware	Battery	2000mAh – 2600mAh (Optional)
Hardware	USB Port	Type C
Hardware	SIM Tray Type	Nano
Hardware	Shell	Material: ABS, Fire rating: HB, Drop Test: 1.0M
Hardware	Main LCD	4.3 inch Colored Display
Hardware	Merchant Display	2” x 0.5”
Hardware	Keypad	Numeric Keypad
Software	Remote updates	APP, language & Firmware Update
Software	Secure Connectivity	Support SSL/TLS
Accessories	Ac-DC Adapter	AC 90-300V, DC 5V/1A (Optional)
Accessories	USB Cable	1 mtr USB Type C, Supports 1A Charging
Accessories	Certification	BIS

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Product Brochure
DQR 4.3”



Front Side



Back Side

Soundbox Specification :

Category	Item	Specifications
Hardware	Module	High performance Dual core ARM CortexR
Hardware	Memory	8MB
Hardware	Communication	4G (LTE Cat1) or 4G + 2G support (Optional)
Hardware	Bands Supported	LTE FDD: B1/B3/B5/B8, LTE TDD: B34/B38/B39/B40/B41
Hardware	Indicator	1*RGB LED
Hardware	Speaker	Nominal 2W, Peak 3W > (84-87)dB sound output
Hardware	Audio	PA: 3.2W@5V/4Ω/10% THD
Hardware	Key	4*Keys+1*Reset Key
Hardware	Battery	2000mAh – 2600mAh (Optional)
Hardware	USB Port	Type C
Hardware	SIM Tray Type	Nano
Hardware	Shell	Material: ABS, Fire rating: HB, Drop Test: 1.0M
Hardware	Main LCD	4.3 inch Colored Display
Hardware	Merchant Display	2” x 0.5”
Software	Remote updates	APP, language & Firmware Update
Software	Secure Connectivity	Support SSL/TLS
Accessories	Ac-DC Adapter	AC 90-300V, DC 5V/1A (Optional)
Accessories	USB Cable	1 mtr USB Type C, Supports 1A Charging
Accessories	Certification	BIS

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Product Brochure
Dual- Sim Soundbox



Front Side



Back Side

Soundbox Specification :

Category	Item	Specifications
Hardware	Module	High performance Dual core ARM CortexR
Hardware	Merchant Display	2” x 0.5
Hardware	Brightness	500 nits
Hardware	Memory	16+16 MB
Hardware	Communication	4G
Hardware	Bands Supported	LTE FDD: B1/B3/B5/B8, LTE TDD: B34/B38/B39/B40/B41
Hardware	Bluetooth	5.X supported
Hardware	Wi-Fi	Supported
Hardware	Indicator	1*RGB LED and 1 charging indicator (red)
Hardware	Speaker	Nominal 2W, Peak 3W > (95)dB sound level
Hardware	Audio	PA: 3.2W@5V/4Ω/10% THD
Hardware	Functional Key	4*Keys (Function Key, Power Key, Volume Up, Volume Down) +1*Reset Key
Hardware	Battery	2000mAh – 2600mAh (Optional)
Hardware	USB Port	USB Type C high-power delivery up to 240W
Hardware	SIM Tray Type	Nano (2 SIM Tray Included)
Hardware	Shell	Material: ABS, Fire rating: HB, Drop Test: 1.0M
Software	Remote updates	APP, language & Firmware Update
Software	Secure Connectivity	Support SSL/TLS 1.2
Accessories	Ac-DC Adapter	AC 90-300V, DC 5V/1A
Accessories	Certification	BIS

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Product Brochure

All-in-one Soundbox





Soundbox Specification :

Category	Specifications
Types	Table top with Dynamic QR and Static QR
Module	High performance Dual core ARM CortexR
LCD Screen Size	4.3” Coloured Display
Merchant Display	2” x 0.5”
LCD Brightness	600 Nits
Memory	Flash- 8MB and upto 256MB
Communication	4G (LTE Cat1) or 4G + 2G support (Optional)
Bands Supported	LTE FDD: B1/B3/B5/B8, LTE TDD: B34/B38/B39/B40/B41
Indicator	1*RGB LED and 1 charging indicator (red)
Speaker	Nominal 2W, Peak 3W > (95)dB sound level
Audio	PA: 3.2W@5V/4Ω/10% THD
Functional Key	4*Keys (Function Key, Power Key, Volume Up, Volume Down) +1*Reset Key
Keypad	4X4 Numeric Keypad
Battery	2000mAh – 00mAh (Optional)
USB Port	Type C
SIM Tray Type	Nano
Supported Payment modes	Tap, DIP/Chip insert and Dynamic QR
Cards supported	Mastercard, VISA, Rupay, AMEX and more
Shell	Material: ABS, Fire rating: HB, Drop Test: 1.0M
Remote updates	APP, language & Firmware Update
Secure Connectivity	Support SSL/TLS 1.2
USB Cable	USB Type C, Supports 1A Charging cable (1 mtrs)
Ac-DC Adapter	AC 90-300V, DC 5V/1A (Optional)
Certification	BIS

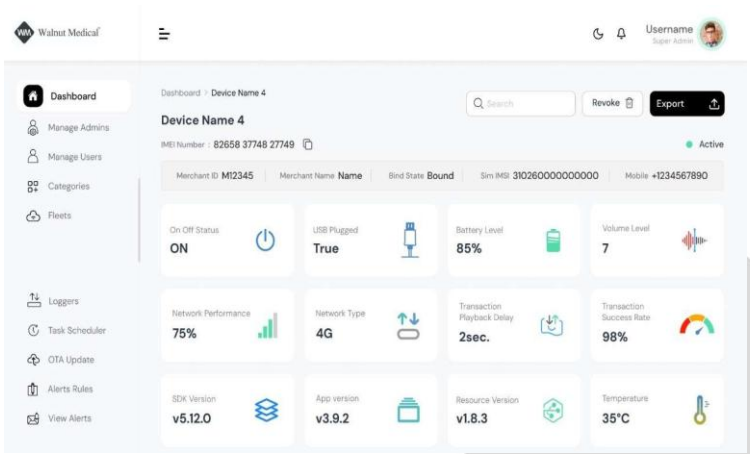
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Product Brochure
TMS + MQTT Broker

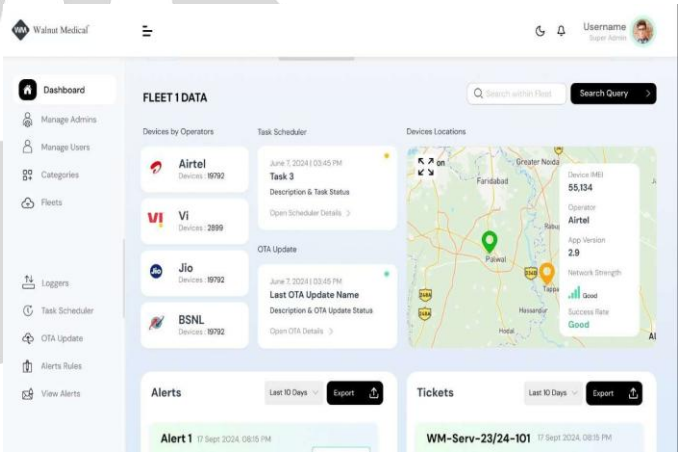
·TMS is a software solution designed to streamline and optimize SoundBox/Terminal operations and management.
·It acts as a control center for terminal fleets, providing real-time visibility and control over terminal activities and configurations.
·TMS streamlines terminal lifecycle management, from initial deployment to ongoing updates and troubleshooting.

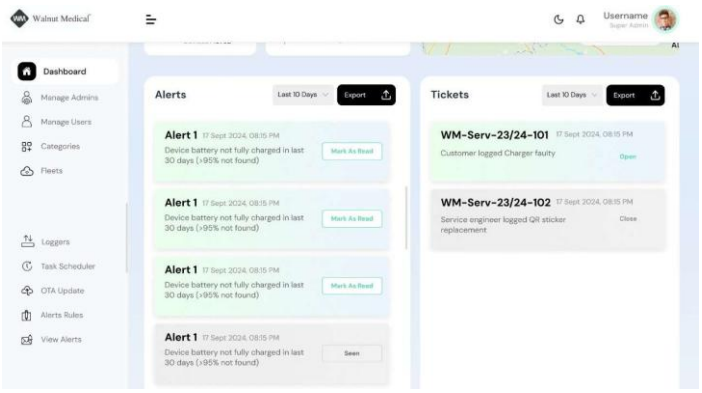


Manage your devices anywhere, anytime
Device and Network info :
Storage usage, memory usage, CPU usage, battery usage, IP address, network strength, IMEI, operator name, device location etc.
APP details :
App version, SDK version, sound file version, language, date of installation, previous version etc.
Scheduler :
Schedule OTA jobs, schedule advertisements, schedule remote commands
Onboarding:
Onboard and link to a merchant account, release OTP, un-board device, monitor transaction matrix.
Defender:
Security metrics monitoring, anomalous behaviour detection, security audits, alerts and mitigation actions

Tracking in One Go

- **Device Model**
- **Device IMEI number**
- **Device Online / Offline status**
- **Device First Time Online Date and Time**
- **Device battery level**
- **Device Audio Volume**
- **Device network strength**
- **Device network type**
- **Device SIM operator**
- **Device SIM details**
- **Device SDK Version**
- **Device App Version**
- **Device File Version**
- **Device Bind State**
- **Device Merchant details (if bind)**
- **Device QR Code (if bind)**
- **Device Transaction Success Rates**
- **Device Transaction Play Delay**
- **Device Switch On Time**
- **Device MCC Code**
- **Device MNC Code**
- **Device IMSI Number**
- **Device Temp**
- **Device USB plugged**
- **Device SIM IMSI**
- **Device Mobile Number**
- **Device task memory**
- **Device heap memory**
- **Device CPU Usage**
- **Device External Memory Size**
- **Device External Memory Used**
- **Device External Memory Free**
- **Device Time Zone**





1. **Payment Trigger Mechanism**
While webhooks are the most commonly used type, the web application can also support WebSockets, Server-Sent Events (SSE), Message Queues, and Pub/Sub Systems.
2. **Device Configuration and Load Balancing on AWS**
Proper configuration and load balancing of devices must be implemented on AWS to ensure efficient operation.
3. **Merchant Onboarding Process**
The process for onboarding merchants may vary depending on the specific requirements.
4. **API Integration**
An API is required for software integration to facilitate communication and data exchange between systems.
5. **Database Schema**
A well-defined database schema is essential for structuring and managing the data effectively within the system.

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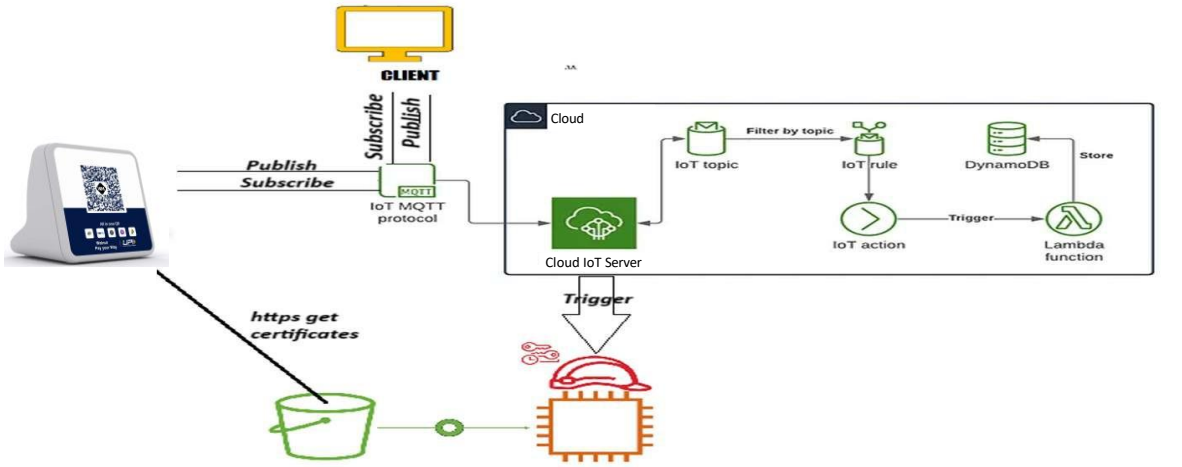
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Product Brochure
TMS + MQTT Broker

MQTT Communication Architecture

- **Efficient Management:** Utilize a single broker to streamline operations and enhance efficiency.
- **Lower Latency:** Benefit from quick transaction announcements, ensuring real-time data processing.
- **Consistency:** Avoid synchronization issues, resulting in fewer failures and smoother operations.
- **Data Security:** Protect your data with robust security measures, ensuring confidentiality and integrity.



Device On boarding - Certs Download from S3-Bucket

Key Features of the MQTT Setup:

Dedicated Connection:
The MQTT communication is isolated from other network traffic, using a dedicated path to ensure low latency and reduced interference from non-MQTT data streams.

Security – TLS 1.2 Encryption:
To protect data in transit, all MQTT messages are secured using **Transport Layer Security (TLS) version 1.2**. This provides:

- End-to-end encryption
- Protection against man-in-the-middle attacks
- Data integrity and confidentiality

Centralized Broker (Non-distributed):
This setup uses a **single, centralized broker** rather than a distributed broker cluster. Benefits of this approach include:

- Easier configuration and maintenance
- Simplified access control and policy enforcement
- Lower complexity in smaller to medium-scale deployments

Not a Distributed Architecture:
The current implementation does not employ a distributed MQTT model. There is no load balancing or horizontal scaling through multiple brokers. This choice is intentional to keep the architecture lightweight and manageable under current scale and reliability requirements.

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